Experiment 1

Student Name: Kushagra

UID:23BAI70105

Branch: BE-AIT-CSE

Section/Group:23AIT_KRG G2

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1. **AIM:** To design and implement the foundational frontend architecture of the EcoTrack application using modern React practices, Vite tooling, and ES6+ JavaScript features.
2. **Objective:**
   a. To set up a React project using Vite with proper project structure
   b. To understand component-based architecture in React
   c. To apply ES6 array methods (map, filter, reduce) for data-driven UI rendering
   d. To separate concerns using components, pages, and data modules

3. **Code: data:**

App.jsx logs.jsx lw.jsx

Eco_Track > src > data > logs.jsx > ...

```
export const logs = [
  { id: 1, activity: "Car Travel", carbon: 4 },
  { id: 2, activity: "Electricity Usage", carbon: 6 },
  { id: 3, activity: "Cycling", carbon: 0 },
  { id: 4, activity: "Bus Travel", carbon: 3 },
  { id: 5, activity: "Solar Energy Usage", carbon: 1 },
  { id: 6, activity: "Flight Travel", carbon: 8 },
];

export default logs
```

Total Carbon function in calculation.jsx

```
dashboard.jsx ×
Eco_Track > src > pages > dashboard.jsx > Dashboard
import Header from '../components/header.jsx'
import logs from '../data/logs.jsx'
import hcarbon from '../data/hcarbon.jsx'
import lcarbon from '../data/lcarbon.jsx'

const sum = logs.reduce((sum, item) => sum + item.carbon, 0);
```

Importing the files in the app.jsx

```
import './App.css'
import Dashboard from './pages/dashboard'
import Header from './components/header'
function App() {
  return (
    <div className="App">
      <Header title="Eco Track: Experiment 1"/>
      <Dashboard/>
    </div>
      )
}

export default App
```

Total Carbon and all activities mapped

```
const sum = logs.reduce((sum, item) => sum + item.carbon, 0);
const Dashboard = () => {

    return (
    <div style = {{
            backgroundColor:'#4CAF50'
        }}>

        <h1>{Header}</h1>
        <p>Total Carbon Footprint: {sum}</p>

        <ul style = {{backgroundColor: 'white',color: 'black'}}>
            {logs.map(log => (
                <li key={log.id}>
                {log.activity}: {log.carbon} kg CO2
                </li>
            ))}
        </ul>
        <p>High carbon emission</p>
        <ul style = {{backgroundColor:'white', color:'red'}}>
            {hcarbon.map(log=>(
                <li key={log.id}>
                    {log.activity}: {log.carbon} kg CO2
                </li>
```

mapping all the footprints with carbon less than or equal to 4 with green

lcarbon.jsx

Eco_Track > src > data > lcarbon.jsx > ...

```
import logs from './logs'
const lcarbon = logs.filter(item => item.carbon <= 3);
export default lcarbon
```

mapping all the footprint with carbon greater than 4

hcarbon.jsx

Eco_Track > src > data > hcarbon.jsx > default

```
import logs from './logs.jsx';
const hcarbon = logs.filter(log => log.carbon >= 4);
export default hcarbon;
```

```
<p>High carbon emission</p>
<ul style = {{backgroundColor:'white', color:'red'}}>
    {hcarbon.map(log=>(
        <li key={log.id}>
            {log.activity}: {log.carbon} kg CO2
        </li>
    ))}
</ul>
<p>Low carbon emission</p>
<ul style = {{backgroundColor:'white', color:'green'}}>
    {lcarbon.map(log=>(
        <li>
            {log.activity}:{log.carbon} kg CO2
        </li>
    ))}
</ul>
```

Output:

Eco Track: Experiment 1

Total Carbon Footprint: 22

- Car Travel: 4 kg CO2
- Electricity Usage: 6 kg CO2
- Cycling: 0 kg CO2
- Bus Travel: 3 kg CO2
- Solar Energy Usage: 1 kg CO2
- Flight Travel: 8 kg CO2

High carbon emission

- Car Travel: 4 kg CO2
- Electricity Usage: 6 kg CO2
- Flight Travel: 8 kg CO2

Low carbon emission

- Cycling:0 kg CO2
- Bus Travel:3 kg CO2
- Solar Energy Usage:1 kg CO2

Learning Outcomes:

- Learnt the use of map, filter and reduce
- learnt how to use the inline css and html inside the return in app.jsx and use it with map and filter
- Learnt the importance of iporting and exporting files and using them inside the app.jsx